

AGENTIC AI LAB
(CSCR3215)
BACHELOR OF TECHNOLOGY
In
Computer Science and Engineering



SHARDA
UNIVERSITY
Beyond Boundaries



Submitted by:

Sanya Chauhan

(2023334971) CSE-I G1

Submitted to:

Mr. Ayush Singh

Technical Trainer

Department of Computer Science and Engineering

School of Computing Science & Engineering

Sharda University, Greater Noida

Jan-June 2026

Autonomous Research Agent using LangChain and Groq API

INTRODUCTION

Artificial Intelligence (AI) has revolutionized the way information is accessed, processed, and utilized across various domains. With the rapid growth of digital content, conducting research manually has become increasingly time-consuming and inefficient. Traditional methods require users to search multiple sources, filter relevant data, and organize it into a meaningful format, which can be both labor-intensive and prone to errors.

To address these challenges, Autonomous Research Agents have emerged as a powerful solution. These agents leverage AI technologies to automate the process of data collection, analysis, and report generation. By integrating intelligent reasoning with external tools, such systems can significantly reduce human effort while improving the quality and consistency of research outputs.

In this project, an Autonomous Research Agent is developed using LangChain, a framework designed for building applications powered by Large Language Models (LLMs). The agent utilizes the capabilities of a powerful LLM through Groq to understand user queries, process information, and generate structured responses.

The system incorporates multiple tools to enhance its functionality. A web search tool is used to gather up-to-date information from the internet, while Wikipedia serves as a reliable source for factual knowledge. The agent combines these sources, analyzes the data, and produces a well-structured report containing key sections such as introduction, concepts, applications, advantages, challenges, and future scope.

This project demonstrates how AI-driven agents can automate complex research workflows and improve productivity. It highlights the practical application of modern AI frameworks in solving real-world problems and showcases the potential of intelligent systems in academic and professional environments.

OBJECTIVE

The objective of this project is to design and implement an Autonomous Research Agent using LangChain that can automate the process of information gathering and report generation.

The specific objectives include:

- To develop an AI-based system that accepts a research topic as input
 - To automatically collect relevant data from multiple sources such as web search and Wikipedia
 - To analyze and summarize the collected information using a Large Language Model
 - To organize the extracted data into a structured format
 - To generate a detailed and readable research report
 - To reduce manual effort and improve efficiency in research tasks
-

KEY FINDINGS

The developed system demonstrates the following capabilities:

- Automatically collects information using web search (DuckDuckGo)
 - Retrieves factual knowledge using Wikipedia
 - Combines multiple data sources for better accuracy
 - Uses an intelligent workflow to simulate agent behavior
 - Generates structured reports including:
 - Introduction
 - Key Concepts
 - Applications
 - Advantages
 - Challenges
 - Future Scope
 - Conclusion
 - Reduces manual effort in research tasks
 - Produces readable and well-organized outputs
-

CHALLENGES

During the development of this project, several challenges were encountered:

- Frequent updates and deprecations in LangChain APIs
- Compatibility issues between different library versions
- Import errors due to changing module structures
- Groq model deprecations requiring model updates
- Dependency issues in the Colab environment
- Handling errors from external tools like Wikipedia and search APIs

These challenges were resolved by modifying imports, updating packages, and implementing a custom agent logic.

FUTURE SCOPE

The project can be enhanced in the following ways:

- Convert reports into PDF format automatically
 - Add more data sources such as research papers and news APIs
 - Build an interactive user interface using Streamlit or web frameworks
 - Implement citation and reference generation
 - Improve accuracy using more advanced LLMs
 - Enable multi-topic comparison and analysis
-

CONCLUSION

The Autonomous Research Agent successfully demonstrates how Artificial Intelligence can automate complex research tasks. By combining LangChain with Groq API and external tools, the system efficiently gathers, analyzes, and organizes information into structured reports.

This project highlights the potential of AI agents in improving productivity and reducing manual effort in academic and professional research.

SAMPLE OUTPUT 1

Role of AI in Healthcare

Introduction:

Artificial Intelligence (AI) is transforming the healthcare industry by improving diagnosis, treatment, and patient care. It helps analyze complex medical data and enhances decision-making, leading to better patient outcomes.

Key Concepts:

- Machine Learning for data analysis and prediction
- Deep Learning using neural networks
- Natural Language Processing for medical records
- Computer Vision for analyzing medical images

Applications:

- Disease diagnosis using medical imaging
- AI chatbots for patient support
- Drug discovery and development
- Predictive analytics for patient outcomes
- Remote patient monitoring

Advantages:

- Improved diagnostic accuracy
- Personalized treatment plans
- Increased efficiency in healthcare processes
- Better patient experience
- Reduced healthcare costs

Challenges:

- Data quality and availability issues
- Lack of clear regulations
- Cybersecurity risks
- Need for clinical validation
- Lack of trained professionals

Future Scope:

AI will enable precision medicine, telemedicine, advanced medical research, and improved healthcare management systems.

Conclusion:

AI is revolutionizing healthcare by improving efficiency and accuracy. Despite challenges, its future potential is significant and will continue to shape modern medicine.

SAMPLE OUTPUT 2**AI in Smart Cities****Introduction:**

Artificial Intelligence (AI) plays a vital role in developing smart cities by improving urban infrastructure, efficiency, and sustainability. It helps manage resources effectively and enhances the quality of life for citizens.

Key Concepts:

- Use of AI, IoT, and data analytics
- Data collection from sensors and devices
- AI systems for decision-making and automation

Applications:

- Traffic management and smart mobility
- Smart energy and waste management
- Improved government services
- Urban safety and disaster management

Advantages:

- Improved efficiency of city operations
- Better resource management
- Increased sustainability
- Enhanced citizen engagement

Challenges:

- Data privacy and security concerns
- High infrastructure costs
- Need for skilled professionals
- Ethical issues in AI systems

Future Scope:

AI will enable intelligent transportation, smart energy systems, and advanced urban planning for more sustainable cities.

Conclusion:

AI is transforming cities into smarter, more efficient environments. Despite challenges, its future potential is significant for urban development.
